A Novel Self-Adaptive Method in Generating DEM of Riparian Zone With Satellite Images and Corresponding Water Level in a Classification Way

Jianbo Tan, Yi Tang, Guang Zhao, Meng Zhang, Zhuokui Xu, Yunfei Zhang, Bo Wang, Wenwen Gao, and Xinyao Xie

***Abstract*— Fine and new digital elevation model (DEM) of riparian zone is urgently required for hydrological and environmental modeling. With the effects of water fluctuation and submersion, acquiring DEM in the riparian zone is difficult. In this study, a self-adaptive receiver operating characteristic (ROC) method was proposed to generate the DEM of riparian zone with the aid of water fluctuation (called ROC DEM). The water extent acts as an altimeter to measure the altitude with the corresponding water level. The elevation of each pixel was the optimal threshold to determine whether it was covered with water or not. With the fluctuation in water, the time-series labels of land or water and the corresponding water level of each pixel were recorded. Finally, the elevation of each pixel was obtained by acquiring the most optimal classification threshold with the ROC algorithm. The proposed method skillfully transformed the problem of acquiring the elevation of the riparian zone into a two-class classification problem. The riparian zone in the Dongting Lake (DTL) was considered as the study area to test the proposed method with the time-series Sentinel-1 synthetic aperture radar (SAR) images and the corresponding water level. The proposed method is feasible to obtain the DEM and its results are more consistent with the actual topography compared to other DEM products. The values of R^2 between the ROC DEM and GLAS and field altitude reach 0.7 and 0.9, respectively. This article presents an alternate method for acquiring the topography in the riparian zone and tidal flat.**

***Index Terms*— Digital elevation model (DEM), Dongting Lake (DTL), riparian zone, self-adaptive receiver operating characteristic (ROC) method, Sentinel-1 synthetic aperture radar (SAR).**

I. Introduction

RIPARIAN zone is a type of ecological transitional region that is located between the inland water ecosystem and the terrestrial ecosystem, demonstrating characteristics of high ecological fragility, heterogeneity, dynamics, and biological complexity [1]. Owing to the interface effect of the water–land zone, it exhibits significant ecological functions in the composite ecosystem, including maintenance of species diversity, water conservation, stabilization of two adjacent ecosystems, interception of material, and energy flow [2], [3]. Therefore, the riparian zone has aroused great attention in the study of hydrology and ecology [4], [5].

The topography is an important factor that regulates the local environment of the riparian zone and plays an important role in quantifying water storage, water-level fluctuation, and sedimentation rate [6]. The distribution of vegetation is also highly related to the terrain in the riparian zone, and the terrain information is important for environmental management and research [7], [8]. A fine and timely topography is promptly required for obtaining accurate hydrodynamic simulations, such as flood extent, water flows, and storage volume in numerical simulations [9], [10]. The accuracy of the digital elevation models (DEMs) has been regarded as the most important factor that regulates the precision of hydrodynamic simulations [11], [12].

With the aid of remote sensing approaches, the DEMs are available on global scales, such as Global Multiresolution Terrain Elevation Data (GMTED2010) [13], Shuttle Radar Topography Mission (SRTM) [14], Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER GDEM) [15], high-resolution DEM generation from Advanced Land Observing Satellite (ALOS) Panchromatic Remote-sensing Instrument for Stereo Mapping (PRISM) data [16], and WorldDEMTM derived from TerraSAR-X and TanDEM-X missions [17]. These global DEMs are widely used in many fields [9], [10]. However, owing to the water fluctuation in the riparian zone, it is difficult for the conventional method to acquire an appropriate DEM in this

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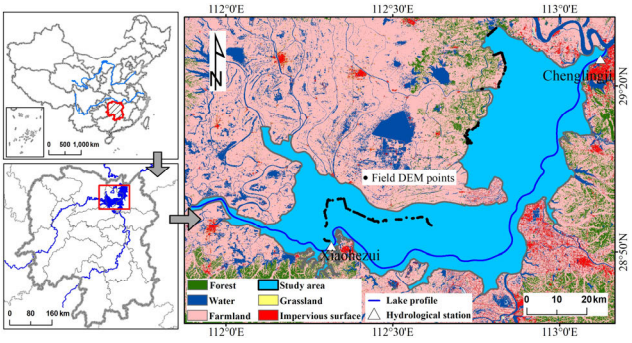


Fig. 1. Location of DTL, hydrological stations, field DEM points, and lake profile.

region [18]. Furthermore, due to the drive of land use and cover change (LUCC) and the hydrological dynamics, the topography in the riparian zone is significantly altered. The DEM of riparian zone should be updated in a timely manner to achieve a good performance in hydrological modeling. Also, finding a feasible way to acquire the DEM in seasonal flooded regions is important and necessary.

The airborne sensors, light detection and ranging (LiDAR), and optical stereo matching are the common way to acquire the finer DEM [19], [20]. However, applying them over large area is time-consuming and costly process [21]. Synthetic aperture radar (SAR) can be used to acquire DEM data through a technique called interferometry. However, the InSAR method is sensitive to the land cover changes and unsuitable in the riparian zone where the water level is fluctuating rapidly [18]. The waterline method was applied to acquire the topography in a widely extended tidal flat with satellite data [22], [23], [24]. During the waterline method, the water line is regarded as an altimeter to measure the altitude of riparian zone [23]. Considering the noise in the SAR or spectral images, there are many misclassifications in water, which further influence the precision of waterline and the precision of DEM. At the same time, many boats floating on the river or lake can also lead to some uncertainties to water line [18]. Also, many works are required to interpolate the void pixels during the acquisition of DEM with the waterline method. A more feasible method that can tolerate these influences should be developed.

In this study, a novel self-adaptive method based on classification way was proposed to acquire the DEM of riparian zone. With the relationship between water extent and corresponding water level, the proposed method acquired the elevation by searching for an optimal threshold rather than waterlines. The proposed method is fault-tolerant and feasible by converting the problem of acquiring the DEM into a classification problem.

Correct information of water extent with high temporal resolution is desirable for the proposed method. SAR images have superiority in retrieving surface waters [25], [26]. The inundated pixels often have a lower backscattering coefficient and appear dark on SAR images for its specular reflection of radar signal [27], [28]. For the 24 h and all-weather acquisition capabilities, SAR is widely applied in mapping the water surface [29]. However, the existence of sidelobes hinders the application of SAR, which distorts SAR images [30]. Also, the registration of time-series SAR images should be conducted to keep spatial consistency among them [31].

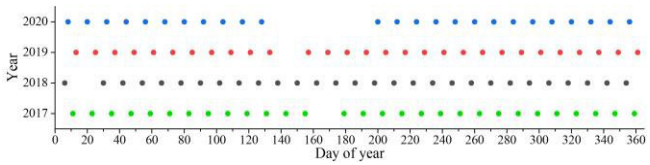


Fig. 2. Dates of Sentinel-1 images taken in the study area.

The proposed method was tested in the riparian zone of Dongting Lake (DTL) in this study. The acquired DEM was further compared with other DEM products, ICESAT-2 ATL08 datasets, and ground field points to identify the accuracy and feasibility of the proposed method. The main objectives of this study were to: 1) transform the obtained DEM into a two-class classification problem; 2) propose a self-adaptive method for acquiring a DEM that can withstand the noise of satellite images; and 3) reduce uncertainties in generating the DEM of riparian zone.

II. Material and Data

A. Study Area

The riparian zone in DTL is the study area to test the feasibility of the proposed method (Fig. 1). The DTL is located in the middle reaches of the Yangtze River. DTL is characterized with flood plain, which provides suitable habitats for rare birds and is listed in the Ramsar List (the List of Wetlands of International Importance) [32]. Its average annual temperature is 17 °C and annual rainfall ranges from 1100 to 1400 mm [33], [34]. Two hydrological stations (Xiaohezui and Chenglingji station) are located at the beginning and end of the study area (Fig. 1). The riparian zone in DTL is broad and its water fluctuation is significant. At the same time, great changes have taken place in its topography for the driver of hydrodynamics and human activities [35].

B. Data Preparation

Sentinel-1A and -1B were launched in one constellation by the European Space Agency (ESA) (on April 3, 2014 and April 25, 2016). Sentinel-1 was equipped with the C-band (5.405 GHz) SAR, which is a part of the Copernicus program [26], [36]. Sentinel-1 has a near-polar sun-synchronous orbit, and the mean local solar time is 18:00 at its ascending node with a 12-day repeat cycle [37]. The satellite images are freely accessible and widely applied in Earth science. Sentinel-1 images have four modes, including interferometric wide (IW)-swath mode, strip map (SM) mode, wave mode (WV), and extra wide (EW)-swath mode [26]. The high-resolution ground range detected production (GRDH) products acquired in IW mode with VH polarization were used in this study [38]. Sentinel-1 images were obtained from the Alaska Satellite Facility (ASF). The 112 Sentinel-1 SAR images within four years (2017–2021) at different water levels were collected to detect the water extent (Fig. 2).

We preprocessed the Sentinel-1 images to acquire the backscattering coefficient of each pixel. The GRDH products

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in IW mode and VH polarization of Sentinel-1 are used in this study with a ground resolution of 20 × 22 m and a pixel spacing of 10 × 10 m in range × azimuth [37], [39]. The SNAP toolkit, including thermal noise removal, radiometric calibration, speckle filtering, Doppler terrain correction, and decibel processing, was used to process data [18], [26]. With aid of the tutorials of Sentinel Toolboxes, all preprocessing was conducted by the SNAP toolbox. To automatically preprocess Sentinel-1 images, the preprocessing workflows were designed using the SNAP graph builder tool and processed in the graph processing tool (GPT) [40].

For the limitation of hydrological stations, only the hydrological data of Xiaohezui and Chenglingji stations are available. Water levels in the dates of Sentinel-1 images were collected. Considering the local solar time of the Sentinel-1 images, the water levels at the two hydrological stations at 18:00 were obtained. These processes maintained the time consistency between water level and Sentinel-1 images.

The ALOS DEM was applied as the original DEM in the proposed method. ALOS DEM was generated by Japan Aerospace Exploration Agency (JAXA) using the archived data of the PRISM onboard the ALOS [16]. These data were used for its high spatial resolution (14 m) and resampled into 10 m using a nearest sampling method to keep consistency with the spatial resolution of Sentinel images.

The ICESat-2 LiDAR datasets (launched in September 2018) were used to further identify the accuracy of the proposed method. The ICESat-2 ATLAS has three laser beams along-track, and the cross-track distance between adjacent beams is 3.3 km [41]. The ICESat-2 ATL08 datasets of DTL from 2018 to 2020 were collected [42], and only the elevation of points uncovered with water was applied to compare with the corresponding elevation derived from the proposed method. The 241 field ground DEM points (Fig. 1), acquired by a GPS receiver in September 2021, were applied to verify the accuracy of the receiver operating characteristic (ROC) DEM in this study. At the same time, both the SRTM digital elevation mode (SRTM DEM) at high spatial resolution (30 m) and the ALOS DEM were spatially compared to the ROC DEM in detail.

III. Methodology

Combined with fluctuation of water level, the elevation of each pixel can be applied to determine whether it was covered with water or not. Therefore, the elevation can be regarded as a threshold to identify the information of inundation along with the water level. With the ever shifting of land and water of each pixel along with water level, we can acquire an optimal threshold that can determine whether the pixel can be covered with water or not, i.e., its elevation. Therefore, the problem of how to acquire the elevation was transformed into a two-class classification problem.

The water level represents the height of the water surface, while the elevation of the DEM represents the height of the land surface. When the water level is below the elevation of the land surface, the land points are not inundated, which are labeled as land (L), while when the water level is higher than the elevation of the land surface, the land points are flooded and are labeled as water (W). With the ever shifting of water level, the labels of one position are correspondingly changing. Also, the height of the land surface can be regarded as the threshold to distinguish between land and water classes (Fig. 3). An optimal threshold of the two-class classification is near the elevation of the pixel. When the optimal threshold of all the pixels is acquired, the DEM of the riparian zone will be available. Therefore, the problem of acquiring DEM was transformed into acquiring the optimal threshold of the two-class classification. The time-series water extent derived from satellite images is used for water–land labeling. In fact, the water level is the hypothetical value for the pixel that is not covered with water in the case of labeled L.

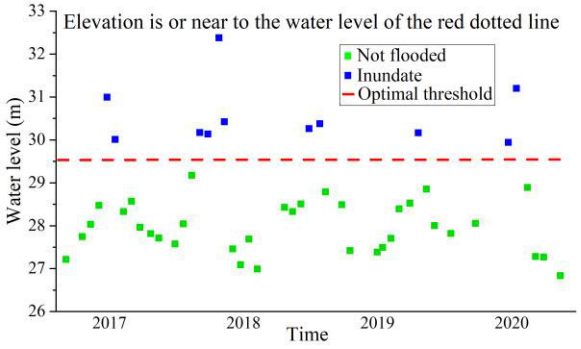


Fig. 3. Conceptual diagram for applying the water level and water extent to acquire the altitude of each pixel.

How to acquire the optimal threshold for imbalanced classification may challenge the traditional classification method. The ROC curve method is a type of moving threshold that acquires the optimal classification threshold from imbalanced datasets [43], which is widely applied for acquiring the optimal probability threshold in machine learning algorithms [44]. The ROC has received a great deal of research interest for model performance assessment, especially for classification [34], which was applied to acquire the optimal threshold.

The workflow of the proposed method is presented in Fig. 4, which mainly includes four steps.

A. Extraction of Water Extent

For SAR images, the open water surfaces exhibit lower backscattering coefficients than the other features due to the specular reflection of the radar signal on the water surface [18]. Therefore, the group of the pixel with a lower value corresponds to water and the other group with a high value corresponds to the land areas (Fig. 5). In this study, we automatically extracted the time-series water extent based on the histogram thresholding of the backscatter values of Sentinel-1 with Otsu algorithms, which is widely applied in automatic classification [28], [45], [46]. The algorithm exhaustively searches for the threshold that minimizes the intraclass variance and maximizes interclass variance, which can be defined in the following equation:

$$\mathrm{TH} = \arg\left\{ \min_t \left(\omega_W(t)\sigma_W^2 + \omega_L(t)\sigma_L^2\right) \text{ and } \max_t \left(\omega_W(t)\omega_L(t)(\mu_W - \mu_L)^2\right) \right\} \quad (1)$$

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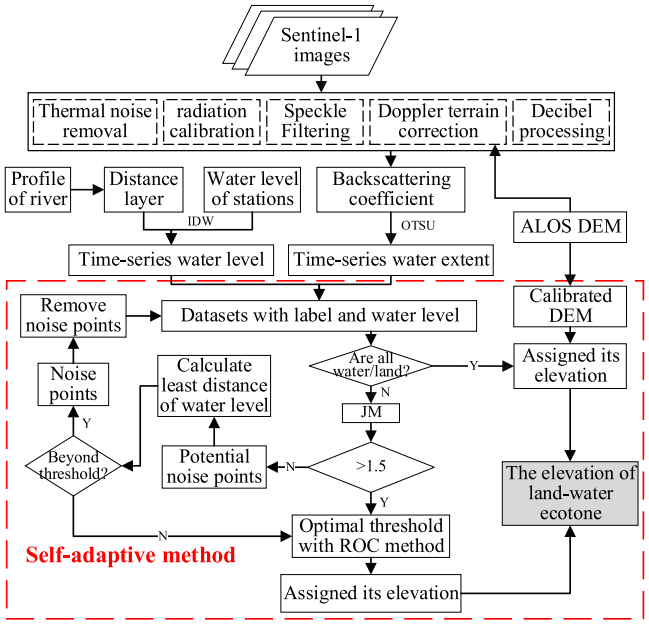


Fig. 4. Workflow of the proposed method.

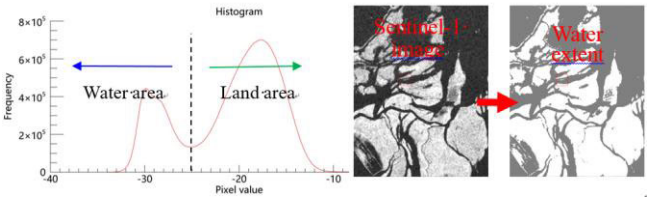


Fig. 5. Histogram of Sentinel-1 backscattering coefficient and water extent extraction with the Otsu algorithm.

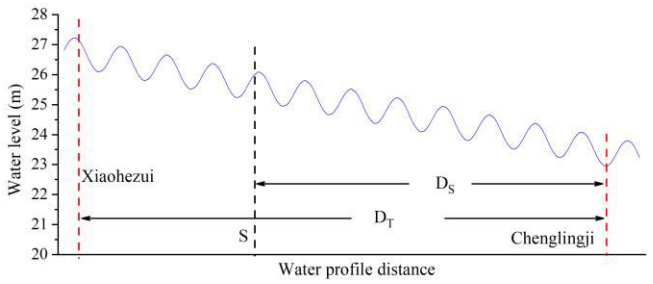


Fig. 6. Water-level profile of DTL.

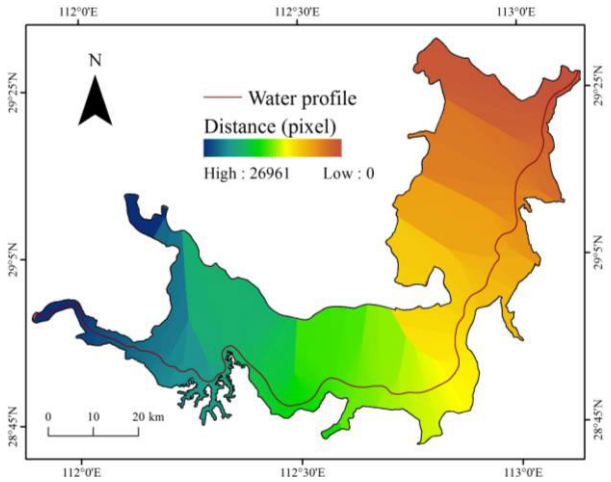


Fig. 7. Profile distance layer of DTL.

where TH is the threshold; ω_W and ω_L are the probabilities of water and land classes separated by a threshold t, respectively; σ_W and σ_L are variances of these two classes, and μ_W and μ_L are the mean values of these two classes.

The Otsu algorithm is a classic histogram thresholding method to determine a threshold by maximizing the variance between two classes, which is the most widely adopted approach used for flood mapping [47]. The Otsu method is less complex and easily operated, making it suitable for large and time-series SAR image scenes (Fig. 5) [48]. With the time-series SAR images, the time-series labels of each pixel are available using the Otsu method.

B. Acquiring the Water Level

In this study, information regarding the water level of Xiaohezui, Chenglingji station, is available. The difference in the water levels of the Xiaohezui and Chenglingji hydrological stations ranges from 0.3 to 8.3 m. The difference in the water potential drives the water flowing, and the elevation of the water surface ranges with the distance (Fig. 6). To acquire the water level of each pixel feasibly, we assumed that the changes in the water level along with the water profile are linearly related, and the inverse distance weighting (IDW) method can be applied to interpolate the elevation of each point. By using the water level of the two hydrological stations and distance layer in this study area, we calculated the time-series water level of each pixel with the aid of the IDW method

$$H_S = H_C + \frac{D_S \cdot (H_X - H_C)}{D_T} \tag{2}$$

where H_X and H_C are the water levels of Xiaohezui and Chenglingji station, respectively; H_S is the water level of S; and D_T and D_S are the profile distances between Xiaohezui and Chenglingji station and between S and Chenglingji station, respectively (Fig. 6).

The line at the center of the lake during the dry season is defined as the water profile line of DTL and the distance of each pixel on the profile line was first acquired [18]. The distance was the length of a line from the Chenglingji station to its position along with the profile line. The distance of other pixels was assigned with the distance of pixel on the profile line that nearest to them, and the whole profile distance of the study area was available (Fig. 7). Coupled with the time-series water level and the profile distance layer, the water level of each pixel in the study area was calculated using the IDW method. By combining with the time-series label and information about the water level, the datasets are arranged for the self-adaptive method.

C. Self-Adaptive Method in Acquiring Elevation

Several scenarios are considered during the self-adaptive process of the proposed method. When the time-series water extent of one pixel is always covered with water or land, the pixel may be permanent water or land (Fig. 8A and

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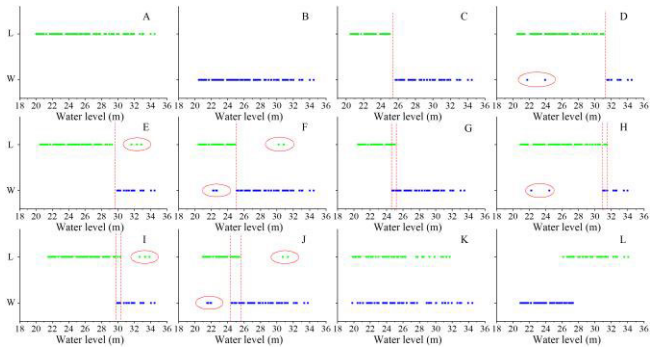


Fig. 8. Scenarios for the relationship between water level and water extent. X axis is water level, green and blue points are the unsubmerged and inundated label, respectively. Red dashed vertical line is the water level near to its elevation.

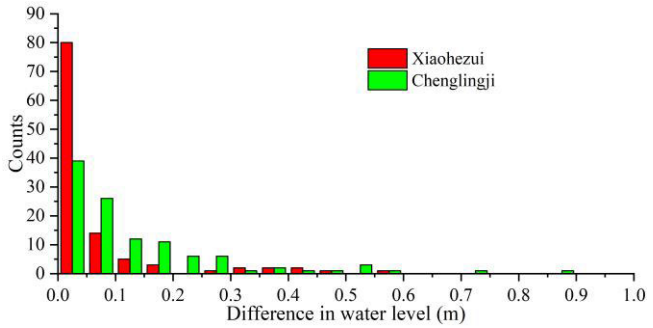


Fig. 9. Histogram of the nearest difference in water level for each hydrological station.

B). When the pixel is temporarily covered with water, the optimal threshold discriminating water and land along with the water level is assigned as its elevation. The Jeffreis–Matusita distance (JM) [(3) and (4)], which is widely applied for determining the complexity for classification tasks, was applied to determine whether the time-series water level with labels can be directly and appropriately distinguished [49]. The statistical distribution in the JM separability criterion has to match the statistical distribution of the data [50]. In this study, a JM value of 1.5 provided a reasonable balance between good separability and minimizing the inclusion of noise with the statistical distribution of the JM value in water level for the binary water and land classes. If the labeled water and land can be directly and appropriately distinguished through a water level, the ROC method can be applied to acquire elevation. Scenario C indicates a time-series water extent without noise, which is completely separable (Fig. 8C). Scenario G indicates a separable in time-series water extent with a narrow overlay in water and land (Fig. 8G)

$$B = \frac{(m_W - m_L)^2}{4 \times \left(\sigma_W^2 + \sigma_L^2\right)} + \frac{1}{2} \times \ln\left(\frac{\sigma_W^2 + \sigma_L^2}{2 \times \sigma_W \times \sigma_L}\right) \quad (3)$$

$$\mathrm{JM} = 2 \times \left(1 - e^{-B}\right) \quad (4)$$

where m_W and m_L are the mean water level of water and land class, respectively, and σ_W and σ_L are variances in water level of water and land, respectively. The value range of separability (JM) is greater than 0 and less than 2.

If the labeled water and land cannot be distinguished, noise points exist, and the process of removing noise points is iteratively conducted (Fig. 4). Scenarios K and L are the extreme cases, indicating difficulty in separating water and land. During the process of removing noise points, the potential noise is initially diagnosed. For each pixel, with the time-series water level and its labels, the points of water may possibly be noise when its water level is lower than the maximum water level of the land and the noise of the water was always located toward the left (Fig. 8D, red circles). On the other hand, the points of land may possibly be noisy when its water level is higher than the minimum water level of land and the noise of land is always located toward the right (Fig. 8E, red circles). Moreover, both noises may occur in the complex study area (Fig. 8F, red circles). The noise also exists in scenario G, which can be presented in scenarios H, I, and J (H, I, and J, red circles in Fig. 8).

For the complexity of scenarios, a least distance indicator was applied to identify the noise points. Theoretically, the difference in the adjacent water levels of the water points or land points is low and within a certain threshold for the continuous change with water fluctuations. Under the influence of noise points, the difference may be abnormal and exceed the threshold. Therefore, the difference in water level can be used to identify whether the point represents noise or not. For example, the red circle in Fig. 8 is too far away from the cluster in the water level, which can be regarded as noise. The water points and land points are ranked with the water level, and the least difference in the water level of each class is obtained

$$\text{least_distance} = \text{neighborhood_dis}(\text{Sort}(\text{waterlevel})) \quad (5)$$

where Sort means ranking the water level of each station.

Theoretically, the distance between the adjacent water levels of each pixel in DTL is similar to that of the hydrological station. The water level in the Chenglingji and Xiaohezui stations can be used to acquire the distance threshold. Based on the histogram of the difference in water level of hydrological stations (Fig. 9), the difference in water level mainly ranges from 0 to 0.9 m. Therefore, the threshold of the difference in the water level (distance) is 0.9 in this study. When the difference in water level is more than 0.9, the point is regarded as noise and will be removed from its time-series data. The procedure of removing noise is iterated until the labeled points can be accurately classified with the optimal threshold (Fig. 4). Using the procedure of removing noise iteratively, scenarios D–F are converted into scenario C, and scenario H–J are converted into scenario G (Fig. 8). This processing is repeated until the labeled water and land can be properly discriminated (Fig. 4).

To guarantee a seamless DEM, a calibrated ALOS DEM is assigned as altitude for scenarios A and B to maintain the consistency between the water level and the ALOS DEM. The water level and ALOS DEM of the points in the waterline near the Chenglingji and Xiaohezui stations were extracted. A linear model between them was established to calibrate the value of ALOS DEM. If the time-series points of the pixel can be distinguished directly and appropriately, the optimal threshold (DEM) can be acquired with the ROC method.

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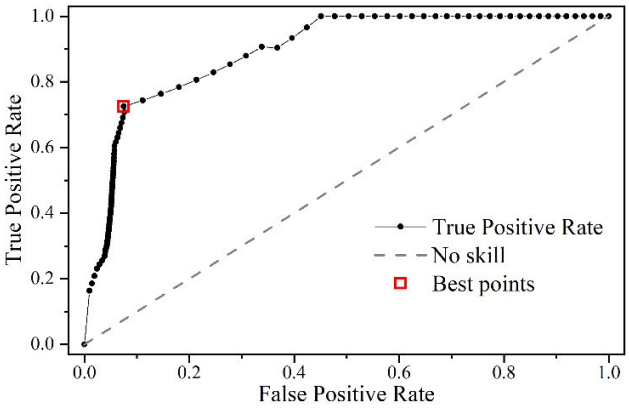


Fig. 10. ROC curve plot for imbalanced classification with the optimal threshold.

The ROC method is a type of moving threshold used to acquire the optimal value from the imbalanced datasets [43], which is widely applied for acquiring the optimal probability threshold in a machine learning algorithm [44]. In this study, the time-series water level is sorted from low to high and deemed as a set of varying "thresholds" used to acquire the ROC curve. When the water level is lower than the "threshold," the points are predicted as land class; otherwise, they are predicted as water class. The predicted labels and actual labels are used to interpret the true positive rate [see (6)] and the false positive rate [see (7)] of the prediction, and the scores are plotted in a line of increasing thresholds to create an ROC curve [49]. Based on the ROC curve, the optimal threshold will be determined with the largest value of G-Mean [see (8) and Fig. 10), which is a metric for imbalanced classification [51]. A point in the top left of the plot indicates a threshold with ideal classification performance (Fig. 10). The average water level between the optimal point and the neighboring point is assigned as the final elevation of this pixel. Using the self-adaptive method, the grid DEM of the riparian zone in the DTL is obtained

$$q(c) = \mathrm{TPR} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \tag{6}$$

$$p(c) = 1 - \mathrm{FPR} = 1 - \mathrm{FP}/(\mathrm{FP} + \mathrm{TN}) \tag{7}$$

$$\mathrm{Optimal_threshold} = \arg\left\{\min_{c}((1 - q(c))^2 + (1 - p(c))^2)\right\}. \tag{8}$$

After obtaining the DEM of the riparian zone, the elevation of the field points, ICESat-2 ATL08, and the other DEM productions were applied for comparison to identify the feasibility of the proposed method in generating the DEM of the riparian zone. The R^2, mean absolute error (MAE), and root-mean-squared error (RMSE) between the generated DEM and the other elevation data were acquired for comparison processing [34], [52].

IV. Result and Analysis

A. Result of the Proposed Method

The DEM obtained by the proposed method is presented in Fig. 11. The closer the position to the center of the lake, the lower its elevation (Fig. 11). On comparison with high-resolution remote sensing images derived from Google Earth in the subwindow, it is revealed that the generated DEM is highly consistent with the actual topography. The boundary of embankment in the DTL is significantly represented in the generated DEM and is highly consistent with the actual case (Fig. 11–1, 2, and 3). For the obtained DEM in DTL, the elevation in the west is generally higher than that in the east (Fig. 11). The ditches and canals are elaborately represented in the generated DEM (Fig. 11–1 and 2). On the whole, the generated DEM demonstrates good agreement with the actual topography.

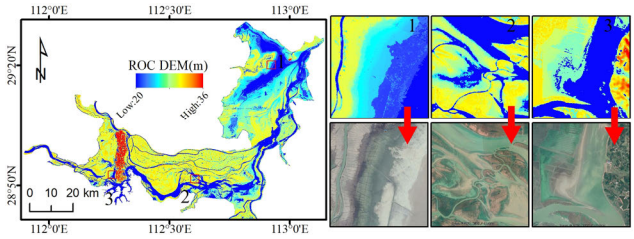


Fig. 11. ROC DEM as well as local detail of subwindow and corresponding high-resolution remote sensing images derived from Google Earth.

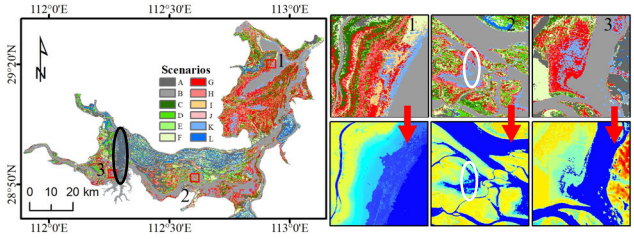


Fig. 12. Spatial map of scenarios, as well as local detail of subwindow and corresponding DEM with the proposed method (the legend of DEM is the same as in Fig. 11).

The map of scenarios generated by the proposed method is presented in Fig. 12. Scenario A is primarily located in the countryside, i.e., the Junshan and Longhu Mountain, where its elevation is high and it is rarely flooded (Fig. 12, black circles). Scenario B is mainly distributed at the center of the lake where the elevation is low, and it is always covered with water. The calibrated ALOS DEM was assigned as the altitude of the region of scenarios A and B. Due to the difference between the ALOS DEM and water level, this may lead to some uncertainties (Figs. 11–2 and 12–2, white circle). Scenarios C and G, which are featured with no noise, are mainly located in the region that is frequently inundated. Scenario K is mainly distributed in the regions close to or in permanent water such as the patches in the middle of rivers, boats in Fig. 12–3. Scenario L is mainly distributed in the regions close to or in permanent land (Fig. 12–2). Combined with the relationship between water level and water extent (Fig. 8), we may conclude that the points of the land labels may possibly be noise for scenario K and the points of water labels may possibly be noise for scenario L. This noise affects the processing of classification, which causes the elevation in scenarios K and L to be inconsistent with that from its surrounding areas (Fig. 12).

Scenarios A and B are the most distributed scenarios in DTL (Figs. 12 and 13), and each of them accounts for more than 10% of the total area. Scenarios C and G, which are featured with no noise, only account for 5.8% and 14.4% of the total area, indicating the presence of an abundance of noise in the

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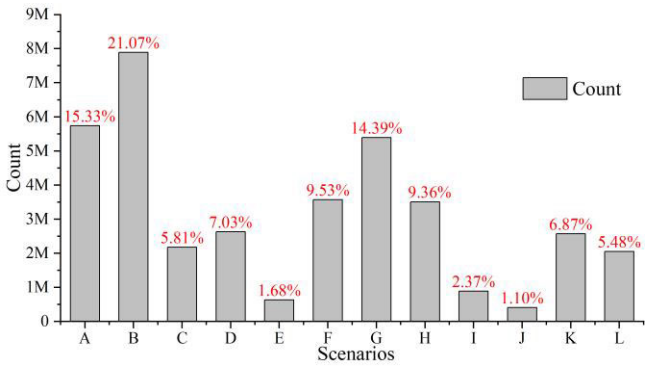


Fig. 13. Number of pixels for each scenario, as well as its proportion of total number.

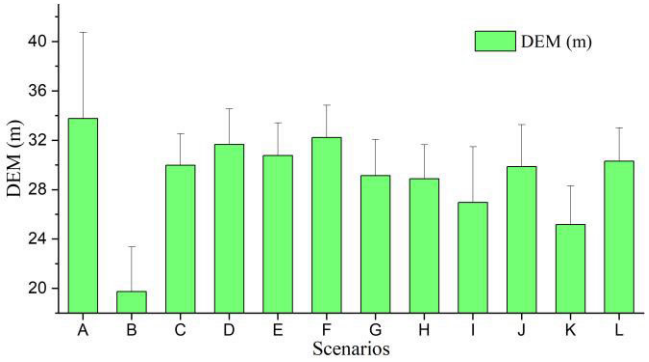


Fig. 14. Mean altitude of each scenario, as well as its standard deviation.

time-series water extent. The number of pixels with a narrow overlay in water and land (scenarios G–I) is larger than the ones without overlay (scenarios C–E). The number of pixels of scenarios D and H is larger than that of scenarios E and I, implying that the noise for the water points is more significant than that of the land points. Scenarios K and F account for more than 12% of the total area.

The mean elevation of each scenario is presented in Fig. 14. Scenario A is featured with the highest elevation. On the other hand, scenario B is featured with the lowest elevation. The elevation of the scenario with a narrow overlay in water and land (scenarios G–J) is lower than that without overlay (scenarios C–F). The elevation of the pixel with noise in water labels (scenarios D and H) is higher than that in the land labels (scenarios E and I). This may possibly be the reason that the pixels in high elevation that are not covered with water are easily misclassified as water for the effects of rain, and the pixels in low elevation that are floated with ships are easily misclassified as land.

B. Comparing to Water Line DEM, SRTM DEM, and ALOS DEM

The water line method is widely applied in acquiring the DEM of the tide flat [22], [23], [24] and the riparian zone of the DTL [18] with satellite data. The spatial distribution of the difference in DEM between the proposed method and the water line method is presented in Fig. 15. Overall, the difference between them is insignificant. The significant difference is primarily located in the ditches, canal, and embankment. The

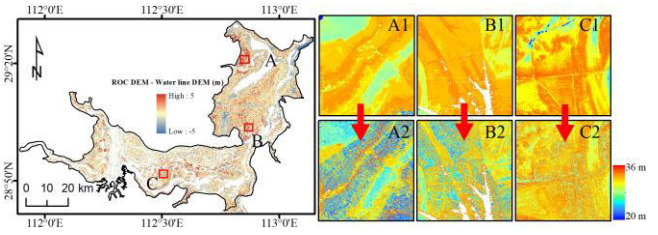


Fig. 15. Comparison between the ROC DEM and water line DEM (ROC DEM subtract water line DEM), as well as local detail of DEM with proposed method (A1, B1, and C1) and waterline method (A2, B2, and C2).

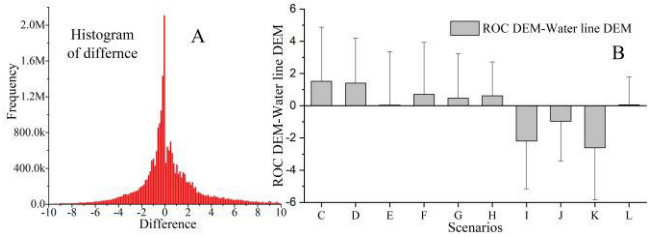


Fig. 16. (a) Histogram of difference and (b) mean difference value between the DEM with the proposed method and the water level method for each scenario, as well as its standard deviation.

overestimated regions are mainly located at a high elevation, while the underestimated regions are close to the permanent water region. For the estuary in the east, there is a significant overestimation in the DEM of the water line due to the effects of the ships. In the northern portion of the study area, there is an immense difference between them (Fig. 15A1). Due to the effects of rain and noise of the SAR images, the elevation of several pixels is underestimated in the water line DEM (Fig. 15A2). This is the same case in subwindow B. In subwindow C, there exist several abnormal low elevation points in the water line DEM, while it is not the case for the DEM with the proposed method. On comparisons with the water line DEM where many abnormal lines are located, it is observed that the DEM derived with the proposed method seems closer to reality.

The difference between the DEM generated by the proposed method and the water line method along with the scenarios and its histogram is presented in Fig. 16. The difference mainly ranged from −2 to 2 m [Fig. 16(a)]. The mean value of the difference is 0.235 m, implying an insignificant difference between them. For scenarios I and K, which have noise in the land labels, the DEM generated by the water line method is significantly higher than that by the proposed method due to the effects of ships [Fig. 16(b)]. In scenarios C, D, and H, which have noise in the water labels, the DEM acquired by the water line method is underestimated by the proposed method due to the effects of rain. Scenarios E–G and L are featured with a lower difference between them. This leads to the conclusion that the proposed method exhibits good tolerance to the effects of noise, rain, and ships.

A stripe in study area was applied to compare the DEM with the proposed method, SRTM DEM, and ALOS DEM (Fig. 17). The DEM with the proposed method presents the topography in detail, which is highly consistent with the image of Sentinel-2 (Fig. 17). There is a road located in the strip that directly

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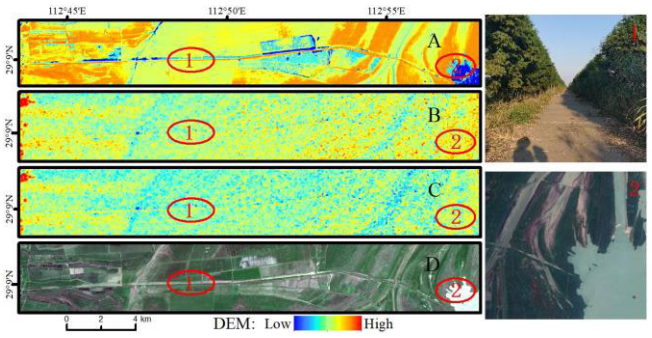


Fig. 17. Comparison among ROC DEM, SRTM DEM, and ALOS DEM, as well as detail images of subwindows. (a) DEM with proposed method. (b) ALOS DEM. (c) SRTM DEM. (d) Sentinel-2 image with the combination of bands 2–4. (1 is the field photograph of subwindow 1; 2 is the image of subwindow 2.)

leads to the center of the lake. The elevation of the road is higher than that of the land on both sides (Fig. 17, subwindow 1). The DEM with the proposed method clearly presents this road, while SRTM DEM and ALOS DEM have difficulty in depicting this. At the end of the road, it is the Changtao Lake, which is always covered with water even during the dry season. Therefore, the elevation in this region is significantly low. Compared with other products of DEM, the DEM with the proposed method exhibits the actual case of topography of subwindow 2 (Fig. 17, subwindow 2). On comparison with SRTM DEM and ALOS DEM, it is observed that the DEM generated by the proposed method is more consistent with the actual topography, implying its good performance.

C. Comparing to ICESat-2 ATL08 and Field DEM

Due to the lack of airborne Lidar data, the transect lines of the ICESat-2 ATL08 of DTL from 2018 to 2020 were applied to verify the accuracy of the DEM produced by the proposed method. The data of four GLAS transect lines during the time of 11/23/2018, 1/7/2020, 2/5/2020, and 12/22/2020 are available in DTL (Fig. 18). When ICESat-2 was overhead, the pixels covered with water were removed, and 3271 points were used for comparison. The footprint size of each photon is 17 m, which is inconsistent with the resolution of the ROC DEM. At the same time, the height datum of the water level derived from the hydrological station is different from that of ICESat-2 ATL08 [53]. Nevertheless, through the scatter plots of the ICESat-2 land elevation of four GLAS transect lines and the corresponding DEM with the proposed method, it is observed that the relationship between them is relatively high and R^2 between them is more than 0.7 (Fig. 18). Comparing to the DEM with water line method [18], the DEM with proposed method is more relevant to the elevation of ICESat-2 ATL08.

The comparison between the ground field DEM and the DEM with the proposed method is presented in Fig. 19. Considering many void values in the water line DEM [18], the water line DEM was not applied for the comparison. R^2 between ROC DEM (DEM with proposed method) and field DEM is 0.9, which is slightly higher than that of SRTM DEM and ALOS DEM, implying a good relationship between ROC DEM and field DEM. The MAE and RMSE, indicating the

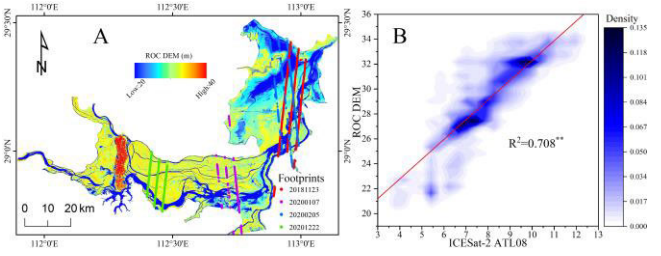


Fig. 18. Comparison transect lines of the ICESat-2 ATL08 and the ROC DEM, as well as their density scatter plot. (a) DEM produced by the proposed method and the location of ATL08 data from 2018 to 2020. (b) Density scatter plot of ICESat-2 elevation and corresponding elevation derived from the DEM produced by the proposed method.

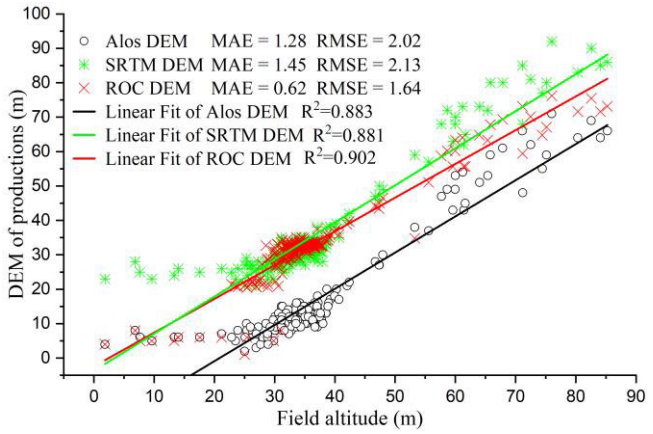


Fig. 19. Scatter plots of field DEM and the DEM productions, as well as their R^2, MAE, and RMSE of linear regression.

residual of the linear regression, of ROC DEM are the lowest. From the comparison, it is observed that the DEM with the proposed method exhibits a better relation with the field DEM than the other DEM productions also, indicating the good performance of the proposed method.

V. Discussion

The water line method has been widely applied in reconstructing the tidal flat topographies of widely extended regions but does exhibit some noteworthy limitations for inland water [18]. To eliminate these effects, a self-adaptive method with an ROC algorithm was proposed in this study to achieve a simpler, robust, and feasible way in obtaining the topography of riparian zone.

A. Superiority of the Proposed Method

The noise of the SAR images and ships floating on the water may greatly affect the accuracy of the water line extraction and further influence the precision of DEM during the process of water line method [18]. To eliminate its effects, the study proposed a self-adaptive method that transformed the problem of obtaining the elevation into a binary classification problem. During the self-adaptive processing, the procedure of removing noise will be iterated until the labeled points can be correctly classified with a threshold. The ROC algorithm, which performs well in imbalanced binary classification, was applied to acquire the optimal threshold. On comparison to the

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DEM with the water line method, the proposed method does not require the extraction of the water line and interpolation of the void pixel [18], and no significant lines appeared. Based on the difference between them, the proposed method greatly eliminates the effects of rain and noise in the high elevation that may be classified into water. Most of the boats floating on the water, which was classified into land, were removed, and little high elevation points appeared along with the water profile. On comparing with SRTM DEM and ALOS DEM, it is observed that the DEM with the proposed method presents the topography in a detailed manner and is more consistent with the actual topography.

The DEM with the proposed method was also compared with GLAS ATL 08 and the field points. R^2 between the DEM with the proposed method and GLAS ATL 08 points is more than 0.7, which is higher than that of the water line method [18]. The DEM points collected from the ground also imply the good performance of the proposed method. Based on the comparisons of the results, the self-adaptive method removed most of the noise, and the ROC algorithm correctly acquired the optimal classification threshold. The method exhibits the ability to withstand the noise points in the water extent. Overall, the proposed method is simple but helpful in acquiring the topography in the riparian zone based on the information of the water level and water extent.

B. Uncertainty Analysis of the Proposed Method

At the same time, we also acquired the difference between the maximum water level of land point (MaxL) and the minimum water level of water point (MinW) for each pixel to present the uncertainties of the water extent. When MinW is larger than MaxL (the difference is greater than 0) and there is no overlay between the water and land labels (scenarios C–F), the larger part of the difference between them implies a longer distance in the water level between the top of the land points and end of the water points, indicating more uncertainty among the water extent of each pixels. When MinW is lower than MaxL (the difference is less than 0), and there may be overlay between water and land labels (scenarios G–J), the larger part of the difference between them implies a longer range overlaid in the water level between the top of the land points and end of the water points also, indicating more uncertainty among the water extent of each pixel. The spatial distribution of MinW–MaxL is presented in Fig. 20. The regions where the difference is significant are mainly distributed in the estuary, canal, and embankment near to permanent water. In these regions, the noise is significant, which may lead to uncertainties in the acquired DEM.

The mean value and the standard deviation of difference between MaxL and MinW for each scenario were presented in Fig. 21. The difference for scenarios C–F is greater than 0 for no overlay in these scenarios. The difference is less than 0 for scenarios G–J for their overlay. The mean value and standard deviation of scenarios C–E are significantly lower than others, indicating its high robustness. The difference in scenarios K and L is significantly high. For scenarios K and L, much of the noise exits (Fig. 8), and it is hard to acquire the optimal threshold.

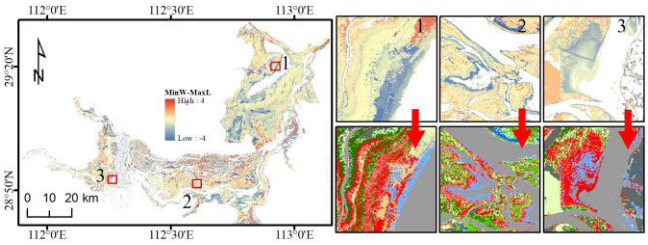


Fig. 20. Difference between maximum water level of land point (MaxL) and minimum water level of water point, as well as local detail of subwindow (the legend of scenario is the same as in Fig. 12).

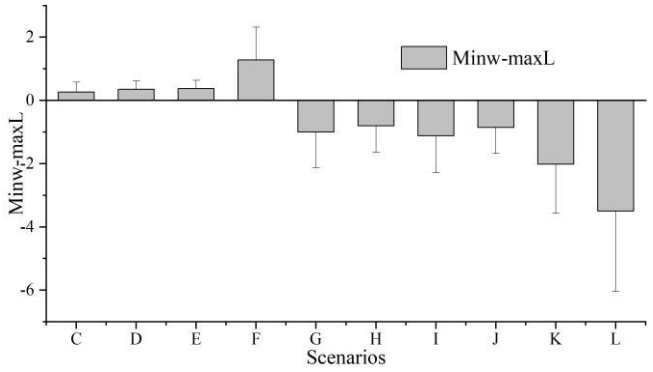


Fig. 21. Histogram of difference between MaxL and MinW for each scenario, as well as its standard deviation.

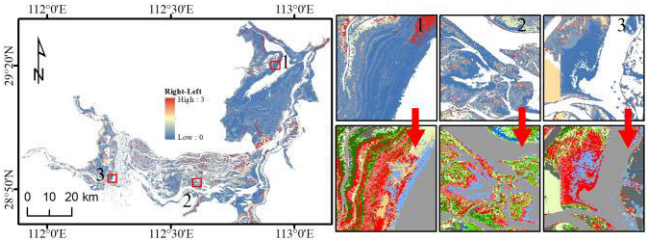


Fig. 22. Gap between the water levels used for the optimal threshold (right water level subtracted from left value), as well as local detail of subwindow (the legend of scenario is the same as in Fig. 12).

During the processing of ROC algorithm, the average water level between the optimal point and the neighboring point was finally assigned as an optimal threshold. The difference in water levels (i.e., the gap) between the optimal point and its neighboring point identifies the uncertainty of the acquired DEM. When the gap is narrow, the threshold derived from the ROC algorithm is closer to the real elevation. To the contrary, the uncertainty is significant when the gap is large. The spatial distribution of gap was presented in Fig. 22. In all, the difference is small, implying the robustness of the DEM with the proposed method. The regions where the difference is significant are mainly distributed in the embankment near permanent water or land. In these regions, the fluctuation of water level is insignificant, leading to a large gap.

The mean value and the standard deviation of the gap between the water levels used for the optimal threshold for each scenario were presented in Fig. 23. The gap of scenarios C–F is larger than that of scenarios G–J. Because of the overlay in scenarios G–J, their gap may be small. Literally removing noise may remove noise points in the time-series dataset, leading to large gap between scenarios F and J.

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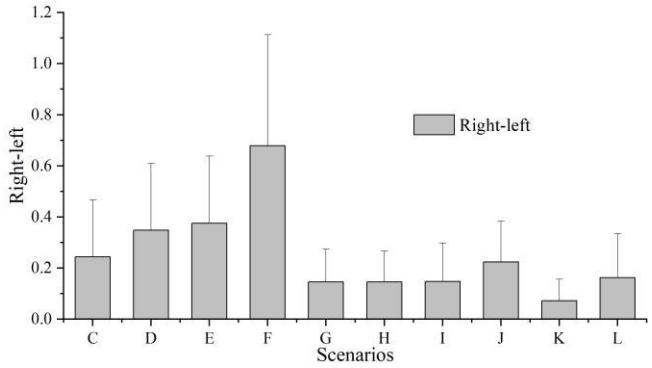


Fig. 23. Histogram of gap for each scenario, as well as its standard deviation.

C. Limitations of the Proposed Method

The proposed method acquires the DEM with the relationship between the water level and water extent, which is close to its altitude rather than its real elevation. This may lead to some difference between the acquired DEM and real elevation. The water extent was acquired by the SAR images, and the water covered with vegetation may be misclassified into land, which may lead to uncertainties in the ROC DEM. For the limitation of field data, only two hydrological stations were applied to acquire the water level of each pixel, which may also induce uncertainties in the water level information.

The linear assumption that changes in water level along with the water profile are linearly related is a simplifying assumption that is commonly used in hydraulic modeling and water resource management [54], [55]. While this assumption may not always hold true in complex hydrological systems with highly variable flow conditions, it is a good tradeoff between efficient and reliability during hydrological modeling.

In theory, the real-time water extent and water level information can achieve a high-precision DEM. For the influence of cloud and rain in DTL, the optical images are almost unavailable, and only the time-series SAR images in recent were applied to acquire the water extent [56]. The time-series data can affect the accuracy of the results by providing more information on the variation in water extent over time. More time-series information can lead to higher precision in DEM with the proposed method. Due to the limitation of satellite images, there are certain pixels in which the number of time-series water extent may not be large enough to accurately acquire its elevation. At the same time, the proposed method may achieve a poor performance in the region where the water fluctuation is insignificant.

The Otsu method was applied in this study to obtain the water extent data. Noise in the SAR images created limitations inherent to the classification method and, in return, caused misclassifications that affected the ROC DEM. Furthermore, the aquatic vegetation can confuse the reflection spectrum of water challenging the precision of water extraction. Due to the sparse distribution of floating aquatic vegetation in the study area, the effects of aquatic vegetation are limited, and the Otsu method can achieve a good performance in distinguishing water bodies [28]. A self-adapting method was proposed in this study to remove the noise and eliminate the effects of misclassification in the water extent. The threshold that determines whether the points are noise or not was derived from the statistics of the difference in the water level of the hydrological stations. The procedure of removing noise may expand the gap in the water level that determines the optimal classification threshold. At the same time, considering the complexity in the DTL, some useful points may be regarded as noise points during the procedure of iteratively removing noise.

Even though there are some limitations, the proposed method performed well on comparison with the DEM derived from the water line method, SRTM DEM, ALOS DEM, GLAS ATL, and ground DEM points. The analysis of uncertainty also presented its feasibility. With the limited data of hydrological stations in some regions, the proposed method provides a simple, robust, and feasible way to acquire the DEM of the riparian zone.

VI. Conclusion

The study presented the high-resolution topographic mapping (10 m) of riparian zone with a novel and simple self-adapting method based on flood-frequency mapping derived from Sentinel-1 images and water level. The method converted how to acquire elevation into a binary classification problem. To eliminate the effect of misclassification or noise points, the iteratively removing noise was induced before classification. The ROC algorithm was applied to acquire the optimal threshold of binary classification for water level. With the result comparison and uncertainty analysis, the DEM with the proposed method performs well. The good performance of the proposed method suggests an alternately feasible and simple method for acquiring the topography in the riparian zone. The thresholds used in this method can be obtained with the statistics of the water level and water extent of each area. This implies the effectiveness of the method in other regions or larger areas where water level and water extent information are available. The DEM we developed represents a fine and recent topography, which can be used for different applications in hydrology, geomorphology, and ecology studies.

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